PLECTA: Overlap-aware instance reconstruction of dense filament networks from binary masks

Oday Allan[a] Stefan Luding[a] Igor Ostanin[a,*]

[a] Multi-Scale Mechanics (MSM), Faculty of Engineering Technology, University of Twente, Enschede, The Netherlands

[To do: Add ORCIDs.]

Abstract

Dense filament networks are difficult to partition into instances because crossings join distinct objects while segmentation gaps divide single objects. We present PLECTA, a deterministic method that converts a binary filament-axis mask into overlap-aware two-dimensional instances. PLECTA skeletonises and decomposes the mask into arms and junctions, alternates exact per-junction continuation matching with globally gated gap matching, and re-estimates tangent, chord, and curvature evidence from the assembled chains. Crossing pixels may therefore belong to more than one output instance.

Evaluation used a method-independent partition of each input skeleton into common fragments. On 128 independently seeded held-out scenes from the same procedural generator, PLECTA achieved mean pairwise F_1=0.854 (density-stratified 95% bootstrap CI [0.845, 0.864]), precision 0.868, recall 0.843, recovery 0.799, and adjusted Rand index 0.851, with no failed scenes. Mean F_1 decreased from 0.911 at 20% coverage to 0.775 at 60%, identifying dense crossings as the main remaining regime of error. Against a greedy minimum-turning-angle continuation rule on identical masks, PLECTA gained +0.152 F_1 [+0.134, +0.169] over 50 scenes, and +0.478 over a released implementation run from its own source at a development-selected configuration. Development-only ablations showed that chord-turn evidence, gap bridging, and junction-topology consolidation made the largest contributions.

CNT-sheet scanning electron microscopy provides the application case. An optional upstream synthetic-data-driven U-Net supplies the binary mask; SEM width measurement and ribbon rendering are downstream characterisation stages and did not affect the held-out grouping result. The evidence supports same-generator synthetic generalisation and a development illustration on real SEM data, not general real-image or three-dimensional reconstruction claims.

Keywords: filament networks; instance segmentation; graph matching; geometric reconstruction; carbon nanotubes; scanning electron microscopy.

[Igor 1] I must say I'm not quite happy with FilaSeg - we need to come up with a better name of the software. I think we do more than just "Seg", I'd avoid that in the name anyways. I would not go towards something trivial like FIBER or STRING either - not enough distinctive. I find attractive something like REUSS, OCCAM or COSSERAT - with reference to physics/mechanics besides the meaning in abbreviation. But that's your baby of course, your word will be the last.

[Response] Addressed. The method is now PLECTA, after Latin *plecta*, a braid or twisted rope. It carries a meaning tied to what the method does, avoids *Seg*, and is not generic. No expansion is forced onto the letters.

*Corresponding author: `i.ostanin@utwente.nl`

1 Introduction

[Igor 2] The paper definitely should have one (and only one) focused, bright central idea. My preference - we present a black box that magically transforms microphotographs of entangled bundles of sticky elastic rods into high-resolution, segmented 3D models. We can frame it as the story about CNTs and make it carbon-specific or talk about arbitrary systems (spagetti, straw, bone tissue etc) and make it an image analysis paper. But we certainly should not pick separate elements of the pipeline and discus only those - this way the value of the paper becomes incrementally small.

[Response] Open decision, not yet settled. The single central idea here is *recovering individual filament identity through crossings*, and the paper is built around it end to end: micrograph, mask, instances, geometry. Two constraints pull against the wider framing. The reconstruction is projected and two-dimensional, so *3D models* would overclaim unless we add depth evidence; and the upstream segmenter is a separate contribution with its own evaluation, so folding it in would mean defending two methods at once. The current draft therefore keeps the CNT story as the application and the identity problem as the idea. Please say if you would rather we widen it.

The properties of carbon-nanotube (CNT) sheets, yarns, and composites depend on the geometry of their bundle networks: length, width, orientation, curvature, and contact topology influence mechanical and electrical response [1–5]. Scanning electron microscopy (SEM) resolves this projected network, but routine image analysis often stops at coverage or orientation distributions [6, 7]. Instance-resolved analysis instead requires each visible bundle to remain addressable across gaps, touches, and crossings.

This requirement is distinct from semantic segmentation. U-Net and its self-configuring variants can identify filament foreground [8, 9], but a binary foreground mask does not encode object identity. A faded segment splits one filament into several connected components; a crossing joins different filaments into one component. The ambiguity is greatest in dense networks, where several local continuations can be geometrically plausible. Compact-object instance priors—bounded blobs, centre-directed flows, or star-convex shapes—are poorly matched to long open curves that may overlap [10–12].

Ridge and vesselness filters enhance curvilinear signal but do not assign identity across junctions [13–15], and fibre extraction packages such as FIRE, CT-FIRE, SOAX, FiNTA and DiameterJ report network statistics rather than resolved instances [16–20].

Recovering individual filaments through crossings is itself established prior art, and the design space is well explored. Basu et al. localise centreline particles, build a minimum spanning tree, delete every bifurcation node, and recombine the resulting segments with a binary integer program over chains of at most three segments, each priced by the integrated squared curvature of a fitted spline; singleton variables let a segment stay unlinked, and the program is re-solved after refitting until the segment set stops changing [21]. DeFiNe consumes a pre-extracted weighted graph and solves a filament cover as a linear program minimising path roughness, with a mode in which one edge may be covered by several paths, so overlaps are representable [22]. SIFNE excises each crossing pixel together with its neighbourhood, estimates a tip direction from the terminus to the local centre of mass, and pairs termini greedily in priority order inside a fan-shaped search region under angular and continuity gates, leaving unmatched tips as filament ends [23]. De et al. resolve crossovers by partitioning a directed graph [24]. Wegmayr et al. group skeleton branches by lowest mutual angle as a baseline and learn pixel embeddings for tangled microplastic fibres, sharing junction pixels in principle but assigning them arbitrarily [25]. DNAi detects junctions morphologically, clusters them by single linkage, erases a disk the width of the fibre, estimates endpoint angles over a short trace, and enumerates the possible pairings at junctions of degree three or four [26]. GraFT builds a graph from a Frangi-filtered skeleton, derives an angle threshold per component, and traces filaments by constrained depth-first search and a greedy longest-first path cover, additionally tracking them over time [27]. Learned alternatives include orientation-aware terminus pairing [28], FibeR-CNN, and recurrent pick-and-trace models [29, 30].

Skeleton graphs, junction excision, continuation angles, distance gates, combinatorial optimisation, iterative refitting and overlapping filament output are therefore all prior art, and none is claimed here. Two distinctions are narrower and survive checking against the sources. First, the methods above either pair endpoints greedily in priority order [23, 25, 27] or enumerate the pairings of a junction only up to degree four [26]; PLECTA instead solves each junction of any degree as one maximum-weight matching in which leaving an arm unmatched is an explicit option at a fixed price, rather than a parity leftover. Second, the methods that optimise globally require intensity evidence or a pre-extracted weighted graph [21, 22], whereas PLECTA reads a binary mask alone. The closest single prior method is that of Liu et al., which likewise works from a mask, reconnects across gaps and permits overlapping output, but pairs termini greedily and depends on a trained orientation network [28].

CNT-specific workflows increasingly combine semantic segmentation with algorithmic tracing and transverse intensity profiles [31]. The present work focuses on the difficult middle step of that workflow: binary mask to overlap-aware instances. This separation is deliberate. It allows grouping to be evaluated independently of the upstream segmenter and prevents width rendering from changing an identity metric. An optional CycleGAN/U-Net route supplies masks for the SEM application [32, 33], but it is not the principal contribution.

We introduce PLECTA, a deterministic graph-matching method for dense filament networks, named after the Latin *plecta*, a braid. PLECTA decomposes a skeleton into arms, endpoint stubs, and merged junction nodes. It then alternates exact per-node continuation matching with globally gated gap matching while tangent and curvature estimates expand from local arms to assembled chains. Its output is a set of projected 2-D instance layers in which crossing pixels may be shared. We evaluate it on 128 unseen scenes from the development generator, on 50 paired clean and degraded scenes, and on two annotated CNT-sheet SEM fields.

2 Materials and methods

2.1 Scope and workflow

The formal input is a binary filament-axis mask $M \in \{0,1\}^{H\times W}$. PLECTA reconstructs an overlap-aware collection $\{I_1, \ldots, I_N\}$, where $I_n \subseteq M$ and a crossing pixel may belong to several instances. The evaluated grouping stage uses only M. A grayscale SEM image is optional downstream input for width and brightness measurement; it cannot revise the fixed grouping. Figure 1 separates this core task from upstream mask extraction and downstream rendering.

2.2 Synthetic data and evaluation split

Procedural scenes preserve the ordered centreline $\mathcal{C}_k$ and rendered support Ω_k of every filament k. Individual supports remain separate, so several filaments may own the same crossing pixel. The generator provides an undegraded one-pixel axis mask, a degraded thin mask containing small gaps and irregularities, a combined thick foreground, an SEM-like image, and the exact overlap-aware instance reference. The degraded one-pixel axis is the input evaluated throughout; it and the reference are shown in Figure 2b,c. Areal coverage is the union area of the thick supports divided by image area; it is not the occupancy of the thin input mask, and Section 3.6 shows it is a proxy for difficulty rather than a cause of it.

Development used 84 scenes: 20 initial scenes across 20–60% coverage, 48 additional scenes at 20, 30, and 60%, and 16 scenes at 40%. The held-out set comprised 128 different seeds: 32 scenes at

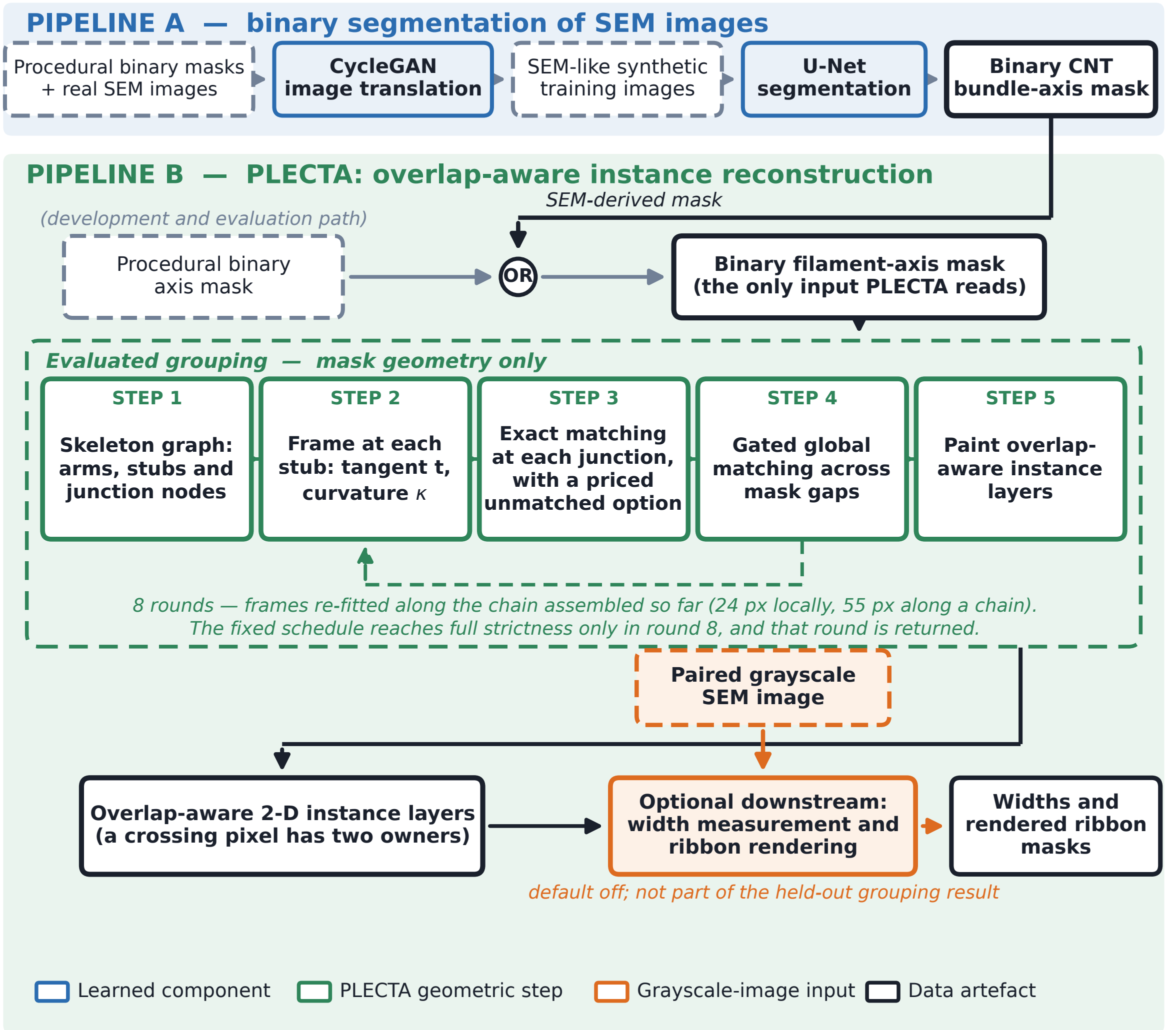

Figure 1. Where PLECTA sits. Pipeline A is the upstream binary segmentation of SEM micrographs; it is not evaluated here. Pipeline B is PLECTA. Its only input is one binary filament-axis mask, which may come either from Pipeline A or, as in every experiment in this paper, directly from the procedural generator. Steps 1–5 operate on mask geometry alone; steps 3 and 4 are re-made over eight rounds against frames re-fitted along the chains assembled so far, and the fixed annealing schedule reaches full strictness only in round 8, which is the round returned. Width measurement and ribbon rendering are downstream, default-off, and require the grayscale image; they contributed nothing to the held-out grouping result, which measures the dashed green box and nothing to the right of it. Colours: blue, a learned component; green, a PLECTA geometric step; orange, a dependency on the grayscale image; dark, a data artefact.

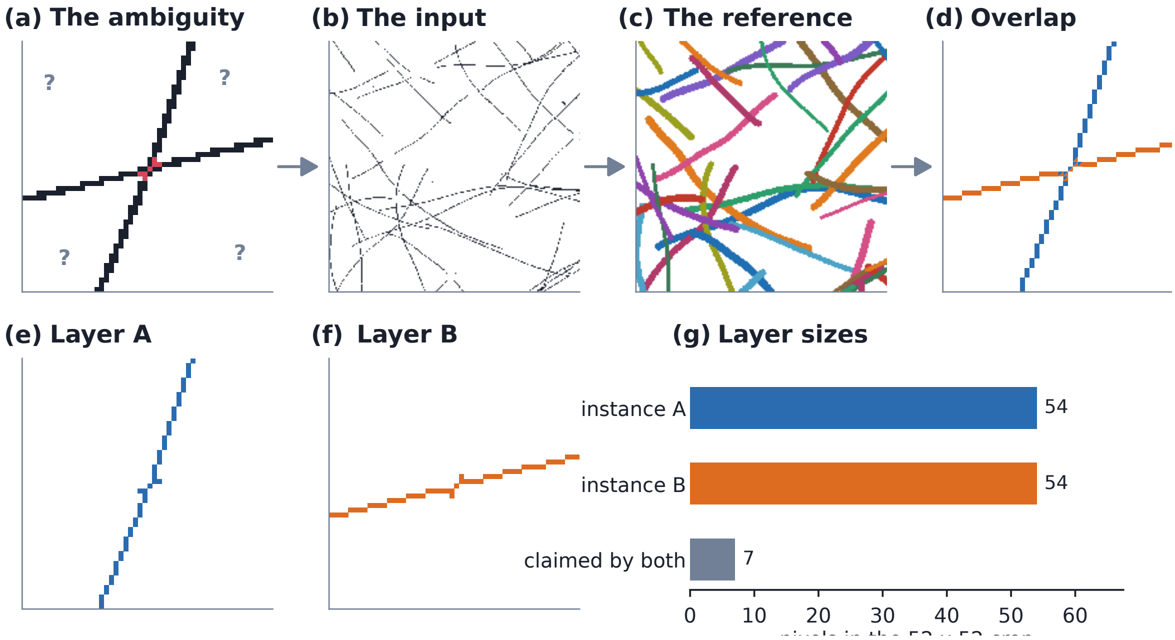

Figure 2. The task, and what the output is. (a) A crossing is one connected blob of mask: four arms leave a junction cluster (red) and nothing local says which continues into which. (b) The whole input actually evaluated — one degraded 1 px binary axis mask, no grey levels and no reference. (c) The overlap-aware reference for the same scene, 40 instances, colours arbitrary and meaningful only within the panel. (d) The crossing of (a) after PLECTA. (e, f) The two instances separately: both contain the junction cluster. (g) Inside this 52×52 crop each instance holds 54 pixels and 7 of them belong to both, drawn as split squares in (d). The output is a set of layers, not a partition, so a crossing pixel may legitimately have two owners.

each of 20, 30, 40, and 60% coverage. It is an independent sample from the same generator, not an independent distribution. A separate 125-scene locked set remains unevaluated. The compact seed manifest records every seed block and an executable verifier checks counts, disjointness, and available scene metadata.

A separate controlled robustness set contained 50 scenes, ten at each of 20, 30, 40, 50, and 60% coverage. Each geometry was scored once from its clean axis and once from its degraded mask. This set was used only to quantify input-degradation sensitivity after the configuration was fixed.

2.3 CNT SEM data and mask sources

Two manually annotated CNT-sheet SEM fields were available for exploratory analysis: B58_100 (1024×1536 px, 280 filaments) and B58_110 (1024×1536 px, 430 filaments). The nominal horizontal field width is 1.66 µm; project metadata associate it with 1536 px, or approximately 1.08 nm px^{-1}. The annotation is one human interpretation of projected, locally ambiguous crossings. Both fields informed development and neither is a population-level test.

Two inputs were derived per field. The first was the skeleton of the manual instance annotation and is therefore an oracle-input condition, not an independent segmentation. The second was predicted by an upstream nnU-Net trained on paired synthetic images and masks; CycleGAN translation is one route from procedural masks to SEM-like training images [32, 33]. B58_100 appeared in that model's training data, whereas B58_110 was verified absent from the relevant training set. Consequently, only B58_110 provides a training-clean observation of mask transfer, and $n = 2$ precludes inference.

Mask quality was compared after skeletonisation with a symmetric 3-px tolerance, avoiding a reward for mask thickness. Reconstruction scores were computed separately within each mask's visible common-fragment population; because those populations differ, cross-mask F_1 values are descriptive rather than paired estimates.

Author metadata required before submission: SEM instrument and detector, accelerating voltage, working distance, specimen preparation, CNT material identity, annotation protocol, and confirmation of the 1536-pixel calibration against the stored image dimensions. The upstream model archive must also supply its architecture, training split, and optimisation settings.

2.4 PLECTA reconstruction

2.4.1 Skeleton graph and local frames

PLECTA binarises M at > 0, skeletonises it, and iteratively removes spurs no longer than 3 px. Skeleton pixels with degree at least three form junction clusters, and adjacent clusters are merged into nodes. Connectors up to 5 px are absorbed into one node; connectors up to 14 px merge nearby nodes but remain real arms; free debris up to 2 px is absorbed. Each remaining connected path is stored as an ordered arm with one stub at each end.

A stub frame contains its tip, outward tangent, and signed curvature. These are estimated by an arclength polynomial fit and are considered reliable only with at least three samples spanning at least 3 px. Local frames use 24 px of support; after links form chains, the support expands to 55 px. Link cost combines direct tangent reversal, the turn from each tangent to the joining chord, and curvature continuity. Gap cost additionally penalises length. Short chords fade the chord-turn term rather than allowing an unstable angle to dominate.

2.4.2 Candidate gates and exact matching

Each of eight rounds first resolves junctions and then gaps (Figure 4). At a junction, candidates must share the node, use different arms, and have finite frames and cost. Exact maximum-weight partial matching selects disjoint pairs while allowing unmatched stubs. With the fixed unmatched price, a full-strictness junction edge is eligible only when its cost is below 1.24.

Gap candidates are constructed only from still-unmatched, reliable stubs. They must belong to different arms and different junctions, lie within 85 px, have direct tangent mismatch at most 0.40 rad, and have chord-turn mismatch at each end at most 0.28 rad. Global exact matching then accepts only edges whose cost is below 0.60 at full strictness. Cycles are broken after junction matching and again after gap matching. The unmatched prices and angular gap limits are annealed linearly from 85% to full strictness. Because only the eighth round is fully strict in the fixed schedule, that round is returned; there is no effective competition among several full-strictness rounds.

2.4.3 Instance output

Connected components of accepted arm links define instances. PLECTA paints every arm pixel and every junction cluster touched by a chain into that instance, so intersecting instances share crossing pixels. An isolated one-arm component shorter than 6 px is omitted. Accepted gap links determine identity but are not painted into the fixed centreline output. All effective parameters, including values previously inherited as code defaults, are materialised in the fixed JSON configuration.

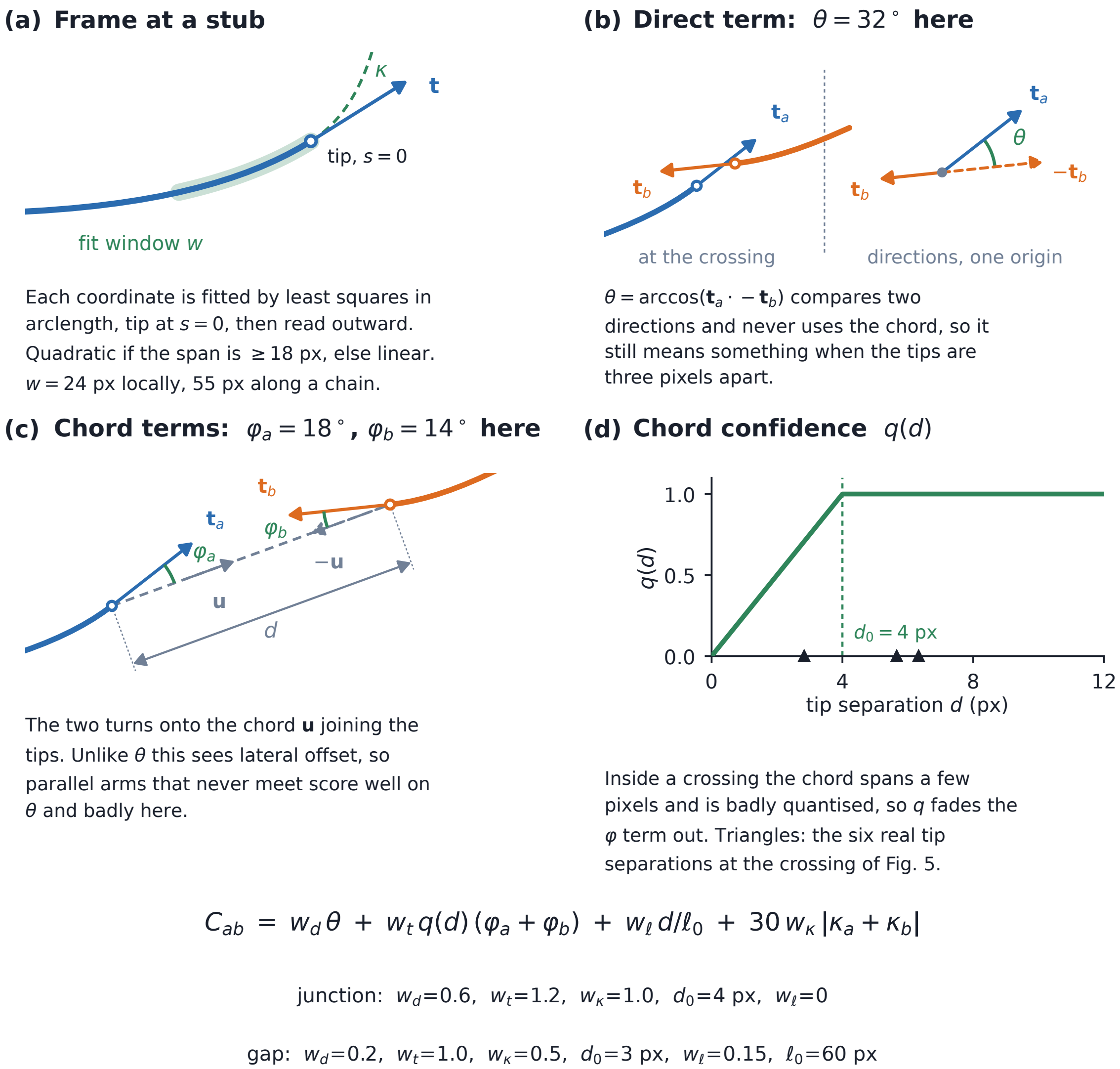

Figure 3. The stub frame and the four terms of the pairing cost. (a) Each coordinate is fitted by least squares in arclength over the window w with the tip at $s = 0$ and then read outward, giving the outward tangent $\mathbf{t}$ and the signed curvature κ; the fit is quadratic when the span is at least 18 px and linear otherwise, and w is 24 px locally and 55 px once the stub can be measured along an assembled chain. (b) $\theta = \arccos(\mathbf{t}_a \cdot -\mathbf{t}_b)$ compares two directions and never uses the chord, so it still means something when the tips are three pixels apart; the reversed tangent is drawn both where it lives and again from a common origin, which is where the angle is read. (c) φ_a and φ_b are the two turns onto the chord $\mathbf{u}$ joining the tips, measured at their own vertices; unlike θ they see lateral offset, so two parallel arms that never meet score well on θ and badly here. (d) Inside a crossing the chord spans a few badly quantised pixels, so $q(d)$ fades the φ term out; triangles mark the six real tip separations at the crossing of Figure 5. The angles drawn in (b) and (c) are stated numerically in the panel titles so the arcs can be checked against them.

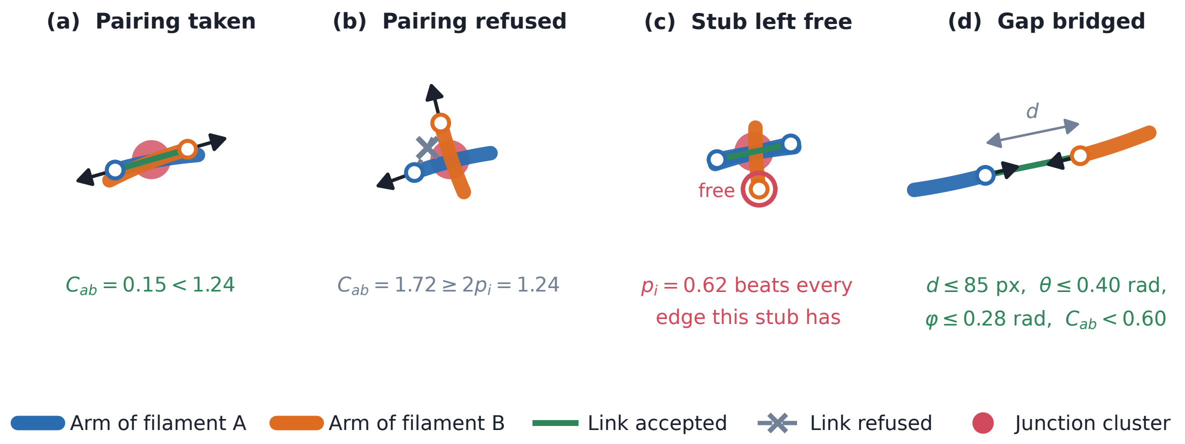

Figure 4. Every decision the linker can make. (a) Two stubs at a junction are paired because their cost falls below the admissibility limit $2p_i$. (b) A competing pairing at the same node is refused: its cost exceeds $2p_i = 1.24$, so the edge is never offered. (c) A stub is left unmatched because paying the price $p_i = 0.62$ scores better than any edge it has; this is what lets a filament genuinely end and stops a T-junction inventing a continuation. (d) Two stubs left free by the junction stage are joined across a mask gap, subject to four gates at once. Every stub is either paired or pays p_i; there is no third outcome. At a junction these decisions are made together rather than one at a time (Figure 5), and a gap is only offered to stubs the junction stage left free. Idealised drawing; the measured values are in Figures 3 and 5.

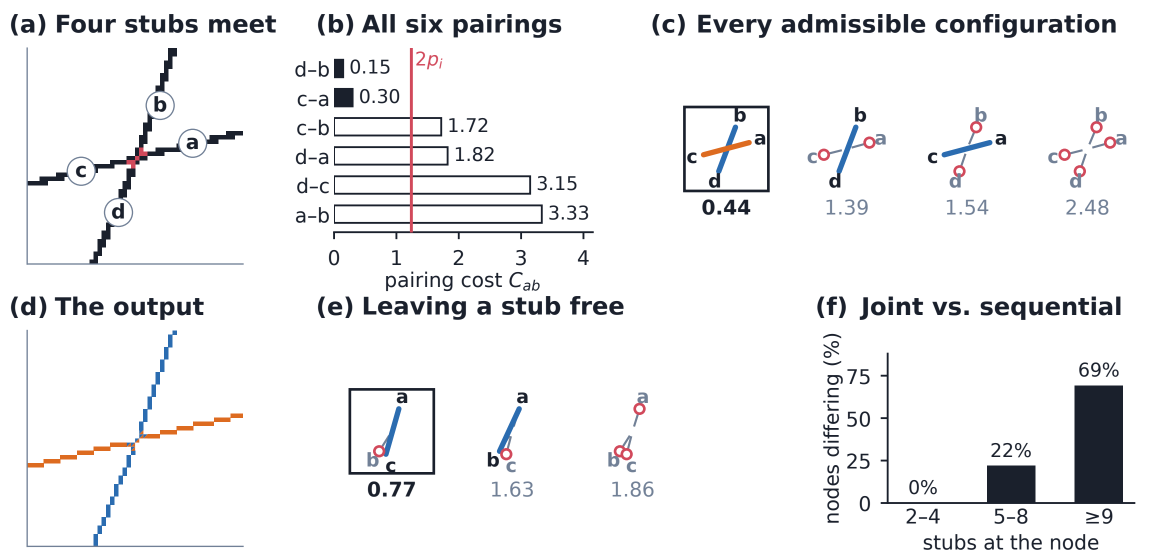

Figure 5. The pairing at a crossing is solved jointly and exactly; every number shown is measured. (a) A real four-arm crossing (development scene `cov20/synth_0001`, node 9). Junction pixels are red; the four arms are still unrelated pieces of skeleton, named a to d anticlockwise by outward tangent. (b) All six pairwise costs. An edge is offered only when $C_{ab} < 2p_i = 1.24$, so four of the six are refused outright (open bars). (c) Every admissible configuration, scored jointly: a configuration scores $\sum C_{ab}$ over its links plus $p_i = 0.62$ for each stub left free, and a maximum-weight matching with edge weight $p_i + p_j - C_{ij}$ minimises that exactly. The boxed configuration is the one returned. (d) The result: two instances sharing the seven junction pixels, drawn as split squares. (e) A real three-arm node (`cov40/synth_0001`). The edge scoring 1.01 is admissible, yet taking it (1.63) scores worse overall than pairing the other two and paying p_i to leave the third free (0.77). (f) Over 20 development scenes the joint solution never disagrees with a sequential cheapest-edge-first rule at a plain crossing ($n = 542$ nodes of degree 2–4), disagrees at 22 % of nodes with 5–8 stubs ($n = 187$) and at 69 % of nodes with 9 or more ($n = 233$), where it is better by a median of 0.90 in total cost. Solving jointly therefore earns its keep in dense tangles, not at clean crossings.

2.5 Optional SEM characterisation and ribbon rendering

After grouping, perpendicular SEM profiles are sampled every 2 px along each ordered chain and at least 6 px from a junction. Profiles are recentred by at most 3 px and accepted only when contrast exceeds both four background-noise standard deviations and 0.05 normalised intensity; fitted widths must lie between 1.5 and 36 px. At least three valid cuts are required. These measurements attach apparent width, brightness, uncertainty, and a quality flag to an existing instance; they cannot alter its arm membership.

Optional ribbon rendering interpolates accepted gaps, resamples the centreline at 1 px, smooths it over 9 px with fixed endpoints, and draws the chain at its fitted width clipped to 3–40 px. It excludes junction blobs and does not clip the ribbon back to the input mask. Silhouette absorption would union rendered instances, but its threshold is zero by default. No absorption or later post-hoc continuation merge contributed to the 128-scene held-out result; neither is presented as part of the evaluated algorithm.

2.6 Metrics and statistical analysis

The primary endpoint is common-fragment pairwise F_1. For each scene, the visible input skeleton is partitioned once into method-independent elementary fragments. Each fragment receives a procedural reference identity and a PLECTA instance identity. A true-positive pair shares both identities; a false-positive pair joins different reference filaments; and a false-negative pair separates fragments of one reference filament. Thus

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \qquad F_1 = \frac{2PR}{P + R}. \tag{1}$$

This is a clustering metric related to the Rand family [34, 35]. It scores grouping, not rendered width, and weights filaments with many fragments more strongly because their pair count is quadratic. We therefore also report adjusted Rand index [36], variation of information split and merge components [37], and the fraction of reference filaments recovered by one dominant predicted instance using 0.8 purity and coverage thresholds.

Held-out means and density strata are accompanied by 95% nonparametric bootstrap intervals (20,000 replicates; seed 20,260,810). Aggregate resampling is stratified by density. The clean-minus-degraded analysis resamples paired scene differences within density. Active-component ablations are one-at-a-time development experiments on all 84 development scenes and are labelled as such; they do not reuse the held-out set.

3 Results

3.1 Held-out synthetic reconstruction

PLECTA completed all 128 held-out scenes with 0 failures. Mean common-fragment pairwise F_1 was 0.854 (density-stratified 95% bootstrap CI [0.845, 0.864]); mean precision, recall, reference-instance recovery, and adjusted Rand index were 0.868, 0.843, 0.799, and 0.851, respectively (Table 1). These scenes used unseen seeds but came from the same procedural generator as development data, so the result measures same-generator generalisation rather than distribution transfer.

Performance decreased with scene density (Figure 6). Mean F_1 fell from 0.911 [0.888, 0.933] at 20% coverage to 0.775 [0.760, 0.790] at 60% coverage; the intermediate 30% and 40% means were 0.874 and 0.858. Recorded per-scene runtime had mean 2.54 s, median 1.12 s, 90th percentile 6.42 s, and maximum 34.06 s; these timings are hardware-specific.

Table 1. Held-out reconstruction scores overall and by target areal coverage. VI denotes variation of information; lower VI is better, whereas higher values are better for the other metrics.

Group	n	F_1	Precision	Recall	Recovery	ARI	VI split	VI merge
All	128	0.854	0.868	0.843	0.799	0.851	0.256	0.216
20%	32	0.911	0.928	0.896	0.918	0.907	0.137	0.093
30%	32	0.874	0.892	0.858	0.828	0.870	0.211	0.161
40%	32	0.858	0.864	0.853	0.796	0.855	0.242	0.224
60%	32	0.775	0.787	0.765	0.656	0.772	0.436	0.388

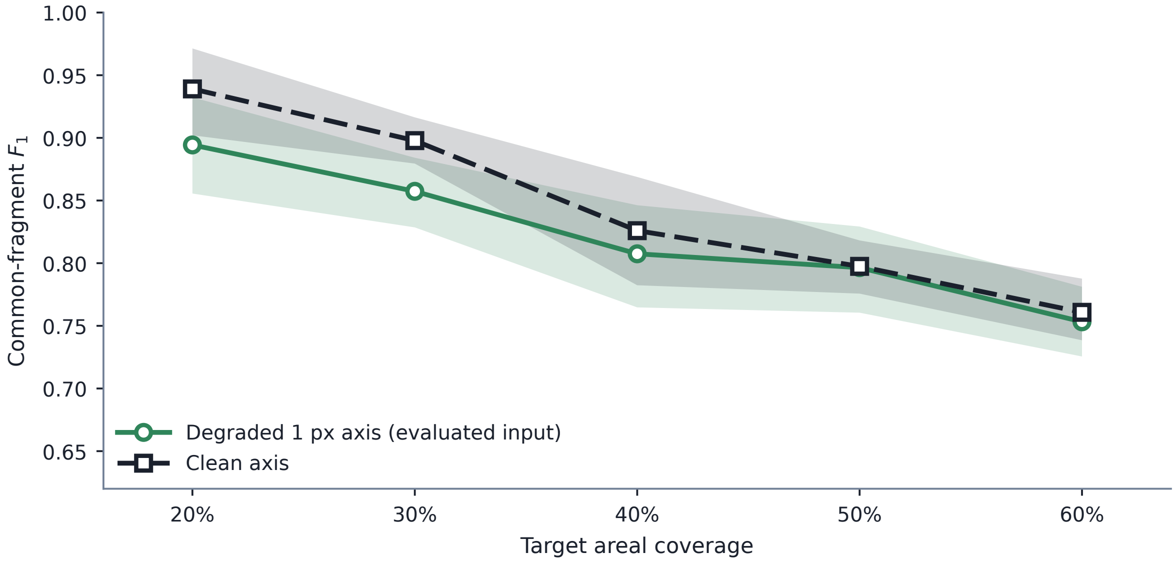

Figure 6. PLECTA against target areal coverage on 50 paired development scenes, ten per coverage level, with each scene scored twice: once from the degraded 1 px axis mask that the paper evaluates and once from the generator's undegraded axis. Lines are means of per-scene common-fragment F_1; the band is a 95 % percentile bootstrap confidence interval for the mean, resampled within each coverage stratum (20 000 replicates, seed 20260811). Degradation costs about 0.045 in F_1 at 20 % coverage and nothing measurable at 50–60 %, where the tangles rather than the gaps dominate. Target areal coverage is the generator setting used to dial in centreline density; it is a proxy for difficulty and not a cause of it, since holding the geometry fixed and varying only bundle width moves realised areal coverage by a factor of 2.49 while F_1 moves by 0.000 (Figure 10b).

3.2 Which density axis matters

Areal coverage is the axis the scenes above are generated and stratified by, but it is not obviously the axis that makes grouping hard. Coverage is the fraction of the frame filled by the rendered bundles, so it rises both when filaments are added and when the same filaments are simply drawn thicker. Only the first changes the grouping problem.

A factorial separates the two. Twelve geometries—three centreline length densities crossed with four seeds—were each rendered at bundle widths of 6, 11 and 16 px. Within one geometry the input axis mask and the overlap-aware centreline reference are byte-identical across the three widths, verified by SHA-256, while the union of the thick supports changes: areal coverage varies by up to a factor of 2.49 within a single geometry. Any movement in the score across those three renderings would mean something in the method or the metric reads the silhouette.

Across all 12 geometries the within-geometry F_1 range was 0.0000, that is, exactly zero in 12 of 12 (Table 2). Areal coverage therefore cannot causally affect the grouping score; PLECTA reads the axis mask, and the metric scores centreline fragments, so neither sees the width the

Table 2. Bundle width against centreline length density, 36 development scenes. Within each row the three bundle widths share one byte-identical axis mask and one reference, so the spread in areal coverage carries no score difference. Length density is centreline pixels per image pixel.

Centreline length density	n	Areal coverage	Bundle width	F_1
0.0193	12	0.118–0.309	6, 11, 16 px	0.920
0.0372	12	0.218–0.505	6, 11, 16 px	0.878
0.0548	12	0.305–0.644	6, 11, 16 px	0.836

Table 3. PLECTA against a greedy minimum-turning-angle continuation baseline on 50 scenes with identical input masks and one scorer. Differences are paired, with density-stratified 95% bootstrap intervals.

Measure	PLECTA	Greedy continuation	Difference
Common-fragment F_1	0.822	0.670	+0.152 [+0.134, +0.169]
Precision	0.831	0.743	+0.088 [+0.070, +0.105]
Recall	0.814	0.615	+0.200 [+0.180, +0.220]
Reference-instance recovery	0.765	0.543	+0.222 [+0.198, +0.246]
Adjusted Rand index	0.817	0.663	+0.154 [+0.137, +0.172]
Variation of information (total) ↓	0.573	1.076	-0.503 [-0.554, -0.453]

bundles were drawn at. Centreline length density does move the score: mean F_1 fell from 0.920 to 0.836 as centreline coverage rose from 0.0193 to 0.0548. The coverage bands of the three length densities overlap—an areal coverage near 0.31 occurs at all three—so coverage does not identify a difficulty level on its own.

Two consequences follow for the rest of this paper. Density strata reported by target areal coverage should be read as a proxy for the length density that accompanies them under this generator, not as a claim about coverage itself. And a mask drawn at a different thickness, as an upstream segmenter may well produce, does not by itself move the grouping score.

3.3 Comparison with greedy continuation

The contribution claimed in Section 1 is that a junction is better solved as one exact matching, with an explicit priced option to leave an arm unmatched, than by pairing endpoints greedily. Greedy priority-ranked pairing is the rule used by SIFNE, by GraFT's path cover, and by the continuity baseline of Wegmayr et al. [23, 25, 27], so the claim is tested directly against it. The comparator skeletonises the same mask, prunes spurs, splits the skeleton into branches, and then greedily accepts the lowest-turning-angle pairing between branch ends within a distance gate of 28 px and a turning-angle gate of 20°. Both methods received identical masks and were scored through the same common-fragment procedure on 50 scenes generated after both configurations were fixed.

PLECTA scored 0.822 against 0.670, a paired difference of +0.152 (95% CI [+0.134, +0.169]), and was better in 48 of 50 scenes (Table 3). Adjusted Rand index and variation of information rank the two methods the same way, so the result does not depend on the in-house measure. The gap is widest in recall and in reference-instance recovery, which is the expected signature: a greedy pass commits to the locally straightest continuation and cannot revise it when that choice leaves a better partner stranded, so single filaments are left divided. On clean axes the difference was +0.158 [+0.144, +0.173], so it is not an artefact of mask degradation. Figure 8 shows the best, median and worst scenes for PLECTA under that contrast.

3.4 Comparison with a released implementation

The greedy rule above is a reimplementation, so we also ran a published method from its own source. DNAi [26] post-processes a binary segmentation: it detects junction pixels morphologically, clusters them by single linkage, erases a disk around each cluster, estimates an endpoint angle over a short trace, and enumerates the possible pairings at junctions of degree three or four. Its segmentation network was bypassed and only this stage was used, so that grouping is not confounded with segmentation quality, and both methods received the same masks and the same scorer.

At its released defaults DNAi scored F_1=0.195 on the degraded masks, which is not a fair measure of it. The cause is its single-linkage clustering of junction pixels at 10 px. Our crossings are closer together than that, so the clustering chains: at 60% coverage the largest cluster spans 89 px and is replaced by one small erasure disk, the crossings along it are never separated, and the whole tangle survives as a single instance. Reducing the clustering distance to 2 px on development scenes raised development F_1 from 0.211 to 0.378 and cut variation-of-information merge from 2.07 to 0.66, confirming the mechanism. Dilating the input to a thicker mask was also tested across 41 development configurations and made DNAi worse at every setting, so it was not used.

At its best development-selected configuration DNAi scored 0.343 against PLECTA's 0.822, a paired difference of +0.478 (95% CI [+0.456, +0.501]) over 50 scenes; on clean axes the difference was +0.414 [+0.392, +0.436]. The gap widens monotonically with density (Figure 7).

This is a limitation of domain and density, not a general ranking. DNAi is built for DNA-fibre micrographs, which are sparse: its own bundled test image contains five junction clusters, whereas our scenes contain between 23 and 183. Where the network is sparse and the axis is clean it recovers 0.660 at 20% coverage. Two structural differences explain the rest. It commits to a pairing at every junction of degree two to four with no cost threshold, so it cannot decline a continuation the way a priced unmatched option can; and it resolves only junctions of degree at most four, though that cap accounts for little here, firing on under 2% of junctions at its shipped setting.

Neither comparison establishes a general ranking. Both are limited to this generator, and Section 4.2 states what remains untested.

3.5 Input-mask robustness

On the separate 50-scene controlled set, mean F_1 was 0.822 from degraded axes and 0.844 from the corresponding clean axes. The paired clean-minus-degraded change was 0.0224 (95% CI [0.0113, 0.0333]). Thus the modelled gaps and raster irregularities caused a small resolved loss under this generator, without testing robustness to a different image distribution. Widening the axis mask costs more than degrading it (Figure 9): a thicker mask skeletonises into a messier graph, which matters because an upstream segmenter controls that thickness.

3.6 Development ablations and downstream rendering

One-at-a-time ablations were evaluated only on the 84 development scenes (Table 4). Relative to the evaluated configuration, removing the junction chord-turn term changed F_1 by -0.1858, removing gap matching by -0.1394, and disabling connector-based junction-cluster consolidation by -0.0634. Removing chain-extended frames and using a single round produced smaller changes of -0.0215 and -0.0199. These development results identify chord-aware continuation, gap matching, and junction-topology preconditioning as the largest tested contributors; they are not held-out effect estimates.

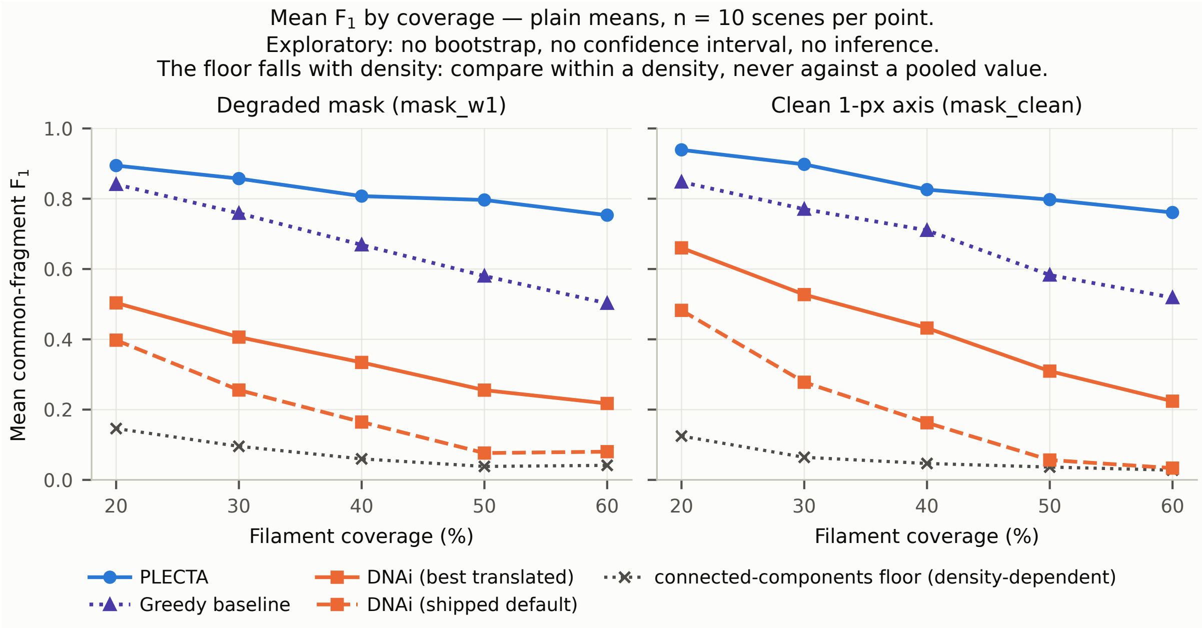

Figure 7. PLECTA, the greedy continuation rule and DNAi across coverage, on the same 50 scenes with identical masks and one scorer, for the degraded and clean axis conditions. Plain means, ten scenes per point, no interval and no inference. The connected-component floor is drawn per density because it is itself strongly density-dependent, falling from 0.146 to 0.041 across this range; a score must be compared with the floor at its own density and never with a pooled value. DNAi is shown at its released defaults and at the best configuration selected on development scenes.

The core output is a grouping, not a set of connected curves. Accepted gap links establish identity but are not painted, so on the development set 0.653 of core instances are delivered as more than one disjoint pixel run and 0.663 contain a branch point. Readers combining PLECTA with a downstream step that assumes one connected curve per instance should account for this.

Default Stage 4 ribbon rendering achieved development F_1= 0.860, a change of -0.0068 from the core centreline output, and reduced those fractions to 0.000 and 0.005. Rendering therefore buys connectivity at a small cost in the identity score, without changing chain membership. Enabling the separate 0.6 silhouette-absorption rule reduced F_1 further to 0.857. Because absorption can merge identities, it remained disabled and neither rendering condition contributed to the held-out grouping result. These topology measurements do not validate apparent width.

3.7 Two-field CNT SEM mask-source case

The two real SEM fields were exploratory development cases (Table 5). From the manual-derived axis, PLECTA reached F_1=0.882 on B58_100 and 0.837 on B58_110. These are oracle-input conditions because the axes were obtained from the manual instance annotations. With the upstream U-Net axes, the corresponding values were 0.438 and 0.452. The U-Net axis had recall/precision 0.782/0.753 on B58_100 and 0.729/0.887 on B58_110.

B58_100 was present in the segmenter's training data; only B58_110 was verified training-clean. Moreover, each input mask defines a different visible common-fragment population, so the manual-derived and U-Net F_1 values are descriptive rather than paired. With two development fields, these observations show the practical dependence on upstream mask quality but do not estimate real-image generalisation.

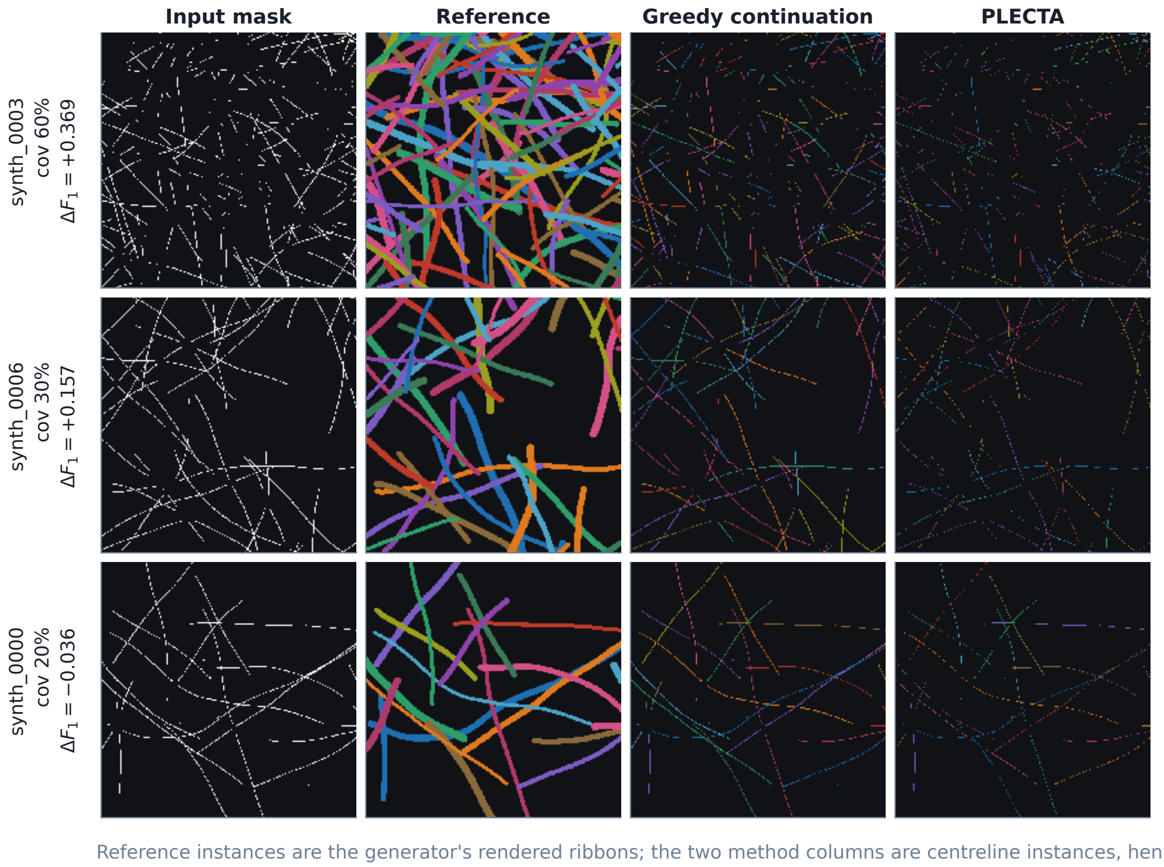

Figure 8. Paired examples against a greedy minimum-turn continuation baseline, on the 50-scene comparison set. Rows are chosen by a stated rule rather than by inspection: the largest PLECTA advantage (`synth_0003`, 60 % coverage, $\Delta F_1 = +0.369$), the scene whose ΔF_1 is closest to the median over all 50 (`synth_0006`, 30 % coverage, +0.157), and the largest comparator advantage (`synth_0000`, 20 % coverage, −0.036). Columns are the degraded input mask, the overlap-aware reference, the comparator and PLECTA. Instance colours are arbitrary and carry identity only within a panel; they are not matched between columns. The reference column shows the generator's rendered ribbons while the two method columns show centreline instances, hence the difference in stroke width.

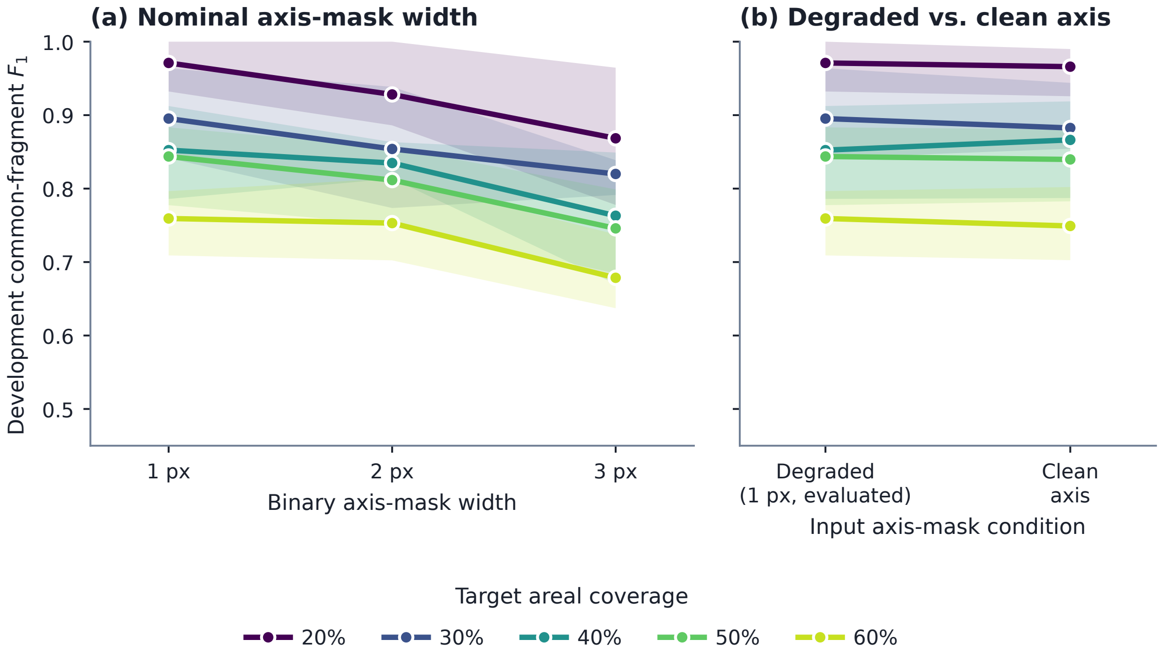

Figure 9. How PLECTA responds to the input mask it is handed, on the development scenes. (a) Nominal binary axis-mask width. (b) The degraded 1 px axis that the paper evaluates against the generator's undegraded axis. Lines are means over the four development scenes at each coverage level and the band is the min–max range across those same four scenes — a description of the spread, not an interval estimate. Exploratory, $n = 4$ scenes per point, no inference. Each variant is scored against common fragments rebuilt from its own input mask, so (a) asks how well the method groups the skeleton it is given rather than how many pixels it was given; the decline with width is the skeletonisation of a thicker mask producing a messier graph, not a change in the grouping problem.

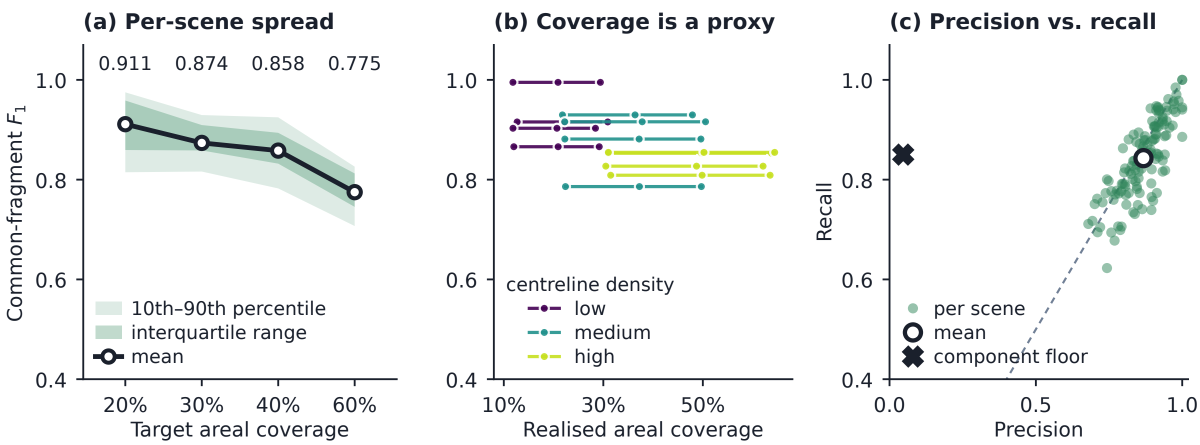

Figure 10. Three properties of the held-out result that a table of means cannot carry. (a) All 128 held-out scenes, 32 per coverage stratum, with the stratum means printed above. The bands describe the spread across scenes and are not an interval for the mean; they overlap heavily, so one scene's score says little about its coverage. (b) Twelve geometries from the development factorial, one line each. Along a line the centreline geometry is fixed and only bundle width changes: realised areal coverage moves by up to a factor of 2.49 while F_1 moves by 0.000. Difficulty follows centreline density, not the amount of ink on the page. (c) The same 128 scenes, one point each. PLECTA sits on the precision–recall diagonal; the one-instance-per-connected-component floor does not, recovering 0.85 of same-filament pairs at a precision of 0.047 because it merges every filament in a connected network into a single instance.

Table 4. One-at-a-time component ablations on the 84-scene development set. Negative ΔF_1 values denote losses relative to the evaluated configuration.

Development variant	F_1	ΔF_1	ARI
Fixed configuration	0.867	+0.000	0.864
No Gap Bridging	0.728	-0.139	0.722
Single Round	0.847	-0.020	0.844
No Chain-Extended Frames	0.846	-0.021	0.842
No Annealing	0.863	-0.004	0.860
No Curvature Term	0.863	-0.005	0.859
No Join_Px	0.845	-0.022	0.841
No Junction-Cluster Consolidation	0.804	-0.063	0.799
No Junction Chord-Turn	0.681	-0.186	0.675

Table 5. Exploratory reconstruction from manual-derived and U-Net axes in two CNT SEM fields. Axis recall and precision compare the U-Net skeleton with the manual-derived skeleton using a symmetric 3-px tolerance.

Field	Input mask	Axis recall	Axis precision	F_1	ARI
B58_100	Manual-derived	–	–	0.882	0.882
B58_100	U-Net	0.782	0.753	0.438	0.435
B58_110	Manual-derived	–	–	0.837	0.836
B58_110	U-Net	0.729	0.887	0.452	0.449

4 Discussion

PLECTA addresses a narrow problem: assigning fragments in a binary filament-axis mask to overlapping projected instances. Its held-out common-fragment F_1 of 0.854 shows that explicit geometry can recover substantial identity without instance-labelled training data or SEM intensity. The mechanism is the combination of local and accumulated evidence. Exact matching at each junction selects mutually compatible continuations, gated gap matching reconnects separated arms, and repeated rounds replace short-arm direction estimates with tangent and curvature estimates from the chains assembled so far. Shared crossing pixels remain in each applicable instance rather than being forced into one label.

The ablations support this interpretation but do not prove that every term is independently necessary. On the development set, removing the junction chord-turn term reduced F_1 by -0.1858, removing gap bridging by -0.1394, and disabling connector-based junction-cluster consolidation by -0.0634. Preventing chain-extended frames produced a smaller decrease of -0.0215. Thus the largest gains arise from solving both kinds of discontinuity—junctions and missing mask segments—with complementary geometric evidence; iterative context provides an additional, more modest benefit. Because these tests reused development scenes, their values describe mechanism and configuration sensitivity, not independent effect estimates.

4.1 Density, masks, and optional rendering

Performance depends on the geometry preserved by the input mask. Mean held-out F_1 declined from 0.911 at 20% coverage to 0.775 at 60%, where crossings and locally plausible alternatives are more frequent. On the 50 paired scenes of the controlled robustness set, generated after the configuration was fixed, the clean axis masks scored 0.0224 higher than degraded masks (95% CI 0.0113– 0.0333). PLECTA can bridge short omissions, but it cannot infer a filament absent from the mask or reliably reject a false axis introduced upstream. Supplying identical masks isolates grouping in an algorithmic comparison; it does not establish end-to-end SEM performance.

Image-informed ribbon rendering is downstream of grouping. On the development set it produced $F_1 = 0.860$, a change of -0.0068 relative to the core output. Its value is connectivity rather than identity: the fraction of instances delivered as more than one disjoint pixel run falls from 0.653 to 0.000, because the core paints arms and junction clusters but not the accepted gap links that carry identity across a break. Silhouette absorption changes the instance partition and scored 0.857; it is therefore optional and should not be conflated with the fixed mask-only grouping result. Width and brightness can be reported for characterised instances, but their physical accuracy requires separate references.

4.2 Evidence boundary

The principal limitation is distributional. The 128 held-out scenes use independent seeds but the same procedural generator family used for development. Their confidence intervals quantify sampling variation within that family, not transfer to another simulator, specimen, microscope, or segmentation pipeline. The ablation and rendering studies are development-only and must remain secondary to the fixed held-out result.

The real SEM study is deliberately a case study with two fields, not a population validation. Manual-derived masks are an oracle-like input condition: they test grouping when the axis evidence is curated, not automatic SEM-to-instance performance. B58_100 is training-contaminated for the U-Net condition, whereas B58_110 is documented as training-clean; only the latter can illustrate an unseen upstream mask, and one field cannot estimate generalisation. Moreover, manual-derived and U-Net masks generate different fragment universes. Their F_1 values therefore describe performance under each input condition but are not a paired comparison of identical objects. More independent fields, sealed before configuration, and annotations from multiple readers are required.

Two further scope limits concern the output itself. First, PLECTA recovers projected two-dimensional identity. Shared pixels encode overlap but not which filament passes above another, so no three-dimensional topology follows. Second, the grouping metric evaluates ownership of centreline fragments and is not a validation of rendered width. Quantitative diameter claims require matched physical measurements across the relevant scale, contrast, and specimen ranges. Pairwise fragment scores also emphasise instances containing many fragments; object recovery and ARI are retained as complementary views.

Both comparisons are bounded. The greedy rule is our reimplementation of a principle, not of any particular paper, and DNAi was run outside the domain it was built for: its own bundled test image holds five junction clusters against 23 to 183 in ours, and part of its cost function reads a colour channel our masks do not have. What the two results support is narrow and mechanical – that pairing endpoints greedily, or committing to a pairing with no option to decline, loses identity in dense crossings – not that PLECTA is generally superior. A ranking would need several released implementations run in their own regimes by their own authors.

An independent procedural benchmark is a practical next test. The `merged_lines` mask generator in the CNT_SEM pipeline can export overlap-aware whole-bundle references and would add a different implementation and morphology model. It should supplement, not replace, the current generator: its field-of-view preset is presently calibrated to a narrow coverage regime, whereas the primary benchmark spans density. A multi-scene run should be made only after pooled calibration is fixed and its seeds are separated from any configuration work. No result from that generator is included in the present evidence. Broader validation should then extend to independent real masks and other filament domains while keeping input fragments and scoring rules fixed.

5 Conclusions

PLECTA reconstructs overlapping filament instances from a binary axis mask by alternating exact junction matching with gated gap matching as local arm geometry expands into chain-level evidence. The fixed method achieved common-fragment $F_1 = 0.854$ on 128 independently seeded held-out scenes from the same procedural generator, with performance declining as network density increased. On identical masks it gained +0.152 F_1 over a greedy minimum-turning-angle continuation rule, the rule shared by several published linkers, and +0.478 over a released implementation run from its own source. Both support solving each junction exactly rather than pairing endpoints in priority order, though the second is a limitation of domain and density rather than a general ranking. Development ablations identify junction chord-turn evidence, gap bridging, and junction-topology consolidation as the principal contributors. Optional image-informed rendering improves geometric presentation but remains separate from the grouping claim. Two real SEM fields demonstrate the workflow under manual-derived and U-Net masks, not population generalisation or width accuracy. The method therefore provides an inspectable, deterministic approach for projected filament identity; broader claims require independent real fields, validated physical widths, a matched comparison against a released implementation, and an external-generator benchmark with a fixed calibration and sealed seeds.

CRediT author contribution statement

[To do: Fill in each author's CRediT roles before submission, checked against actual contributions; keep the author names and order as listed above.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The PLECTA source, complete fixed configuration, compact seed manifest, and aggregate machine-readable evidence are available at `https://github.com/Jorallan/PLECTA`. Generated scenes, detailed per-scene records, and development diagnostics are retained locally. **[To do: Create a tagged archival release and add its DOI before submission; do not infer or preassign a DOI.]**

Acknowledgements

The authors thank the maintainers of NumPy, SciPy, scikit-image, NetworkX, and Matplotlib. The upstream synthetic-data-driven mask workflow adapts the open CycleGAN/U-Net particle-segmentation workflow of Rühle et al. [32]; it is not evaluated as the contribution of this paper.

References

[1] Michael F. L. De Volder, Sameh H. Tawfick, Ray H. Baughman, and A. John Hart. Carbon nanotubes: Present and future commercial applications. *Science*, 339(6119):535–539, 2013. doi: 10.1126/science.1222453.

[2] Jonathan N. Coleman, Umar Khan, Werner J. Blau, and Yurii K. Gun'ko. Small but strong: A review of the mechanical properties of carbon nanotube-polymer composites. *Carbon*, 44 (9):1624–1652, 2006. doi: 10.1016/j.carbon.2006.02.038.

[3] Alexey N. Volkov and Leonid V. Zhigilei. Structural stability of carbon nanotube films: The role of bending buckling. *ACS Nano*, 4(10):6187–6195, 2010. doi: 10.1021/nn1015902.

[4] Igor Ostanin, Roberto Ballarini, David Potyondy, and Traian Dumitrică. A distinct element method for large scale simulations of carbon nanotube assemblies. *Journal of the Mechanics and Physics of Solids*, 61(3):762–782, 2013. doi: 10.1016/j.jmps.2012.10.016.

[5] Bernard K. Wittmaack, Abu Horaira Banna, Alexey N. Volkov, and Leonid V. Zhigilei. Mesoscopic modeling of structural self-organization of carbon nanotubes into vertically aligned networks of nanotube bundles. *Carbon*, 130:69–86, 2018. doi: 10.1016/j.carbon.2017.12.078.

[6] Liyu Dong, Haibin Hang, Jin Gyu Park, Washington Mio, and Richard Liang. Detecting carbon nanotube orientation with topological analysis of scanning electron micrographs. *Nanomaterials*, 12(8):1251, 2022. doi: 10.3390/nano12081251.

[7] Tamjid Imtiaz, Jacques Doumani, Fuyang Tay, Natsumi Komatsu, Stephen Marcon, Motonori Nakamura, Saunab Ghosh, Andrey Baydin, Junichiro Kono, and Ahmed Zubair. Facile alignment estimation in carbon nanotube films using image processing. *Signal Processing*, 202:108751, 2023. doi: 10.1016/j.sigpro.2022.108751.

[8] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*, volume 9351 of *Lecture Notes in Computer Science*, pages 234–241, 2015. doi: 10.1007/978-3-319-24574-4_28.

[9] Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18:203–211, 2021. doi: 10.1038/s41592-020-01008-z.

[10] Carsen Stringer, Tim Wang, Michalis Michaelos, and Marius Pachitariu. Cellpose: A generalist algorithm for cellular segmentation. *Nature Methods*, 18:100–106, 2021. doi: 10.1038/s41592-020-01018-x.

[11] Uwe Schmidt, Martin Weigert, Coleman Broaddus, and Gene Myers. Cell detection with star-convex polygons. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2018*, Lecture Notes in Computer Science, pages 265–273, 2018. doi: 10.1007/978-3-030-00934-2_30.

[12] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3992–4003, 2023. doi: 10.1109/ICCV51070.2023.00371.

[13] Carsten Steger. An unbiased detector of curvilinear structures. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 20(2):113–125, 1998. doi: 10.1109/34.659930.

[14] Alejandro F. Frangi, Wiro J. Niessen, Koen L. Vincken, and Max A. Viergever. Multiscale vessel enhancement filtering. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI'98*, volume 1496 of *Lecture Notes in Computer Science*, pages 130–137, 1998. doi: 10.1007/BFb0056195.

[15] Yoshinobu Sato, Shin Nakajima, Nobuyuki Shiraga, Hideki Atsumi, Shigeyuki Yoshida, Thomas Koller, Guido Gerig, and Ron Kikinis. Three-dimensional multi-scale line filter for segmentation and visualization of curvilinear structures in medical images. *Medical Image Analysis*, 2(2):143–168, 1998. doi: 10.1016/S1361-8415(98)80009-1.

[16] Andrew M. Stein, David A. Vader, Louise M. Jawerth, David A. Weitz, and Leonard M. Sander. An algorithm for extracting the network geometry of three-dimensional collagen gels. *Journal of Microscopy*, 232(3):463–475, 2008. doi: 10.1111/j.1365-2818.2008.02141.x.

[17] Jeremy S. Bredfeldt, Yuming Liu, Carolyn A. Pehlke, Matthew W. Conklin, Joseph M. Szulczewski, David R. Inman, Patricia J. Keely, Robert D. Nowak, Thomas R. Mackie, and Kevin W. Eliceiri. Computational segmentation of collagen fibers from second-harmonic generation images of breast cancer. *Journal of Biomedical Optics*, 19(1):016007, 2014. doi: 10.1117/1.JBO.19.1.016007.

[18] Ting Xu, Dimitrios Vavylonis, Feng-Ching Tsai, Gijsje H. Koenderink, Wei Nie, Eddy Yusuf, I-Ju Lee, Jian-Qiu Wu, and Xiaolei Huang. SOAX: A software for quantification of 3D biopolymer networks. *Scientific Reports*, 5:9081, 2015. doi: 10.1038/srep09081.

[19] Daniel A. D. Flormann, Moritz Schu, Emmanuel Terriac, Divyendu Thalla, Lucina Kainka, Marcus Koch, Annica K. B. Gad, and Franziska Lautenschläger. A novel universal algorithm for filament network tracing and cytoskeleton analysis. *The FASEB Journal*, 35(5):e21582, 2021. doi: 10.1096/fj.202100048R.

[20] Nathan A. Hotaling, Kapil Bharti, Haydn Kriel, and Carl G. Simon. DiameterJ: A validated open source nanofiber diameter measurement tool. *Biomaterials*, 61:327–338, 2015. doi: 10.1016/j.biomaterials.2015.05.015.

[21] Saurav Basu, Chi Liu, and Gustavo Kunde Rohde. Extraction of individual filaments from 2D confocal microscopy images of flat cells. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 12(3):632–643, 2015. doi: 10.1109/TCBB.2014.2372783.

[22] David Breuer and Zoran Nikoloski. DeFiNe: An optimisation-based method for robust disentangling of filamentous networks. *Scientific Reports*, 5:18267, 2015. doi: 10.1038/srep18267.

[23] Zhen Zhang, Yukako Nishimura, and Pakorn Kanchanawong. Extracting microtubule networks from superresolution single-molecule localization microscopy data. *Molecular Biology of the Cell*, 28(2):333–345, 2017. doi: 10.1091/mbc.E16-06-0421.

[24] Jaydeep De, Li Cheng, Xiaowei Zhang, Feng Lin, Huiqi Li, Kok Haur Ong, Weimiao Yu, Yuanhong Yu, and Sohail Ahmed. A graph-theoretical approach for tracing filamentary structures in neuronal and retinal images. *IEEE Transactions on Medical Imaging*, 35(1): 257–272, 2016. doi: 10.1109/TMI.2015.2465962.

[25] Viktor Wegmayr, Aytunc Sahin, Björn Sæmundsson, and Joachim M. Buhmann. Instance segmentation for the quantification of microplastic fiber images. In *2020 IEEE Winter Conference on Applications of Computer Vision (WACV)*, pages 2199–2206, 2020. doi: 10.1109/WACV45572.2020.9093352.

[26] Clément Playout, Yosra Mehrjoo, Renaud Duval, Marie Carole Boucher, Santiago Costantino, and Hugo Wurtele. DNAi: An open-source AI tool for unbiased DNA fiber analysis. *Nucleic Acids Research*, 54(7):gkag335, 2026. doi: 10.1093/nar/gkag335.

[27] Isabella Østerlund, Arne Neumann, Zhiming Ma, Yansong Miao, Staffan Persson, and Zoran Nikoloski. GraFT: A robust network-based spatiotemporal analysis of filamentous structures. *Science Advances*, 12(13):eadz4132, 2026. doi: 10.1126/sciadv.adz4132.

[28] Yi Liu, Abhishek Kolagunda, Wayne Treible, Alex Nedo, Jeffrey Caplan, and Chandra Kambhamettu. Intersection to overpass: Instance segmentation on filamentous structures with an orientation-aware neural network and terminus pairing algorithm. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 125–133, 2019. doi: 10.1109/CVPRW.2019.00021.

[29] Max Frei and Frank Einar Kruis. FibeR-CNN: Expanding Mask R-CNN to improve image-based fiber analysis. *Powder Technology*, 377:974–991, 2021. doi: 10.1016/j.powtec.2020.08.034. Published online 2020.

[30] Yi Liu, Su Peng, Jeffrey Caplan, and Chandra Kambhamettu. Pick and trace: Instance segmentation for filamentous objects with a recurrent neural network. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023*, Lecture Notes in Computer Science, pages 635–645, 2023. doi: 10.1007/978-3-031-43993-3_61.

[31] Torben Peters, John Schumann, Kerstin Kämpf, and Asmus Meyer-Plath. AI-assisted recognition of fibre instances in scanning electron microscope images for aerosol characterization and exposure assessment. *Powder Technology*, 469:121909, 2026. doi: 10.1016/j.powtec.2025.121909.

[32] Bastian Rühle, Julian Frederic Krumrey, and Vasile-Dan Hodoroaba. Workflow towards automated segmentation of agglomerated, non-spherical particles from electron microscopy images using artificial neural networks. *Scientific Reports*, 11(1):4942, 2021. doi: 10.1038/s41598-021-84287-6.

[33] Oday Allan. CNT_SEM: CycleGAN synthesis and U-Net segmentation workflows for carbon-nanotube SEM data. GitHub repository, 2026. URL https://github.com/Jorallan/CNT_SEM.

[34] William M. Rand. Objective criteria for the evaluation of clustering methods. *Journal of the American Statistical Association*, 66(336):846–850, 1971. doi: 10.1080/01621459.1971.10482356.

[35] Ignacio Arganda-Carreras, Srinivas C. Turaga, Daniel R. Berger, Dan Cireşan, Alessandro Giusti, Luca M. Gambardella, Jürgen Schmidhuber, Dmitry Laptev, Sarvesh Dwivedi, Joachim M. Buhmann, Ting Liu, Mojtaba Seyedhosseini, Tolga Tasdizen, Lee Kamentsky, Radim Burget, Václav Uher, Xiao Tan, Changming Sun, Tuan D. Pham, Erhan Bas, Mustafa G. Uzunbas, Albert Cardona, Johannes Schindelin, and H. Sebastian Seung. Crowdsourcing the creation of image segmentation algorithms for connectomics. *Frontiers in Neuroanatomy*, 9:142, 2015. doi: 10.3389/fnana.2015.00142.

[36] Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2(1):193–218, 1985. doi: 10.1007/BF01908075.

[37] Marina Meilă. Comparing clusterings—an information based distance. *Journal of Multivariate Analysis*, 98(5):873–895, 2007. doi: 10.1016/j.jmva.2006.11.013.